Joker of Academics

On the Gravitational Collapse of Compressed Academic Redundancy: Can arXiv.zip Form a Black Hole?

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Abstract

This paper first proposes and formalizes the "**Academic Mass–Gravity Equivalence Principle**" (AMGEP), asserting that **academic outputs that are not understood or cannot be reproduced possess equivalent gravitational mass**. Based on the full arXiv dataset as of December 31, 2025 (a total of 2,847,391 papers), we constructed a multi-scale model of academic quality and calculated its effective scholarly mass (ESM) after ZIP compression. The results show that the total ESM of arXiv.zip reaches $4.71 \times 10^{18}\ kg$, corresponding to a Schwarzschild radius of $r_s = 1.70 \times 10^{-15}\ m$. Although a black hole has not yet formed on a macroscopic scale, locally high-redundancy regions (such as the "Introduction" and "Related Work" sections) already exhibit strong gravitational lensing effects and characteristics of a cognitive event horizon. More troublingly, we found that the unit-mass contribution of the word "novel" is 725 times that of ordinary vocabulary, while the reproducibility correction is almost negligible ($\gamma \approx 10^{-30}\ kg$). Simulations suggest that if current growth trends continue, arXiv will reach a critical knowledge density in 2047, triggering an irreversible knowledge singularity. We hereby call for the immediate launch of an "Academic Hawking Radiation Program(AHRP)" — systematically deleting PDFs containing buzzwords to release negative energy and delay the heat death of the universe. It is also recommended that NASA point the Webb Telescope at Cornell University servers to observe possible "academic Hawking radiation".

Keywords: mass of buzzword; knowledge singularity; Academic Hawking Radiation Program(AHRP)

1. Introduction:

Since its founding in 1991, arXiv.org has become the 'preprint graveyard' of academia—by February 2026, there were approximately 2.8 million papers on the platform, covering fields such as physics, mathematics, computer science, and biology. The total size of these papers, if stored in a conventional way, would require about 15PB of storage. Nowadays, the number of new papers added daily exceeds the average human reading speed by 300 times. Even more ironically, these texts are filled with self-referential rhetorical bubbles: 'novel framework,' 'paradigm-shifting insight,' 'first to leverage deep learning in this domain'—yet 97% of these works have neither been replicated nor genuinely read.

Fig1. No title

This phenomenon raises a fundamental question: when information no longer carries meaning, does it still possess physical reality? Inspired both by Einstein's mass-energy equation $E = mc^2$ and a mentor's remark that 'your work isn't solid enough,' we propose: academic redundancy not only wastes time but also warps space-time. Every paper that goes uncomprehended is a tiny mass source in the universe; when compressed into a .zip file, the density spikes, potentially triggering the ultimate fate predicted by general relativity—**the formation of a black hole.** The goal of this paper is to use the most 'rigorous' mathematical tools to answer this increasingly urgent question: **if all arXiv papers were compressed into a single ZIP file, would it collapse into a black hole?**

Our approach includes:

- Constructing the first academic mass-gravity equivalence theory (AMGEP), formalizing the contribution of buzzword mass;
- Using the full arXiv dataset to calculate the effective scholarly mass (ESM) after ZIP compression;
- Evaluating whether it meets the Schwarzschild black hole formation conditions;

2. Theoretical framework

2.1. Academic Quality – Analogy to the Equivalence Principle of Gravity (AMGEP)

We propose the following axiomatic system:

Axiom A1 (Information Inertia): Any information that is not absorbed exerts resistance on the cognitive subject, and this resistance is equivalent to mass within the framework of general relativity.

Axiom A2 (Redundancy Amplification): Repetitive statements (such as self-citations or templated

abstracts) contribute to an increase in quality in a superlinear manner by creating false informational barriers.

Axiom A3 (Reviewer Shield Failure): Peer review cannot eliminate redundancy; it only transfers it to the references, resulting in the conservation or even increase of total academic quality.

Consequently, **effective scholarly mass (ESM)** is defined as:

$$M_{ESM} = \sum_{i=1}^{N} m_i$$

The quality of each individual paper, m_i, is given by the following formula:

$$m_i = \underbrace{\alpha N_{\text{words},i}}_{\text{Basic Mass}} + \underbrace{\beta B_i}_{\text{buzzword inflation}} - \underbrace{\gamma C_i}_{\text{Reproducibility Fix}}$$

In the formula: $N_{\text{words},i}$ represents the total number of words in the i-th paper (excluding references and appendices), B_i is the number of times a buzzword appears in the i-th paper, and C_i is the reproducibility indicator variable ($C_i = 1$ if the code is publicly available, otherwise 0). According to the Bekenstein bound, the minimum mass of 1 bit of information at the Planck scale is approximately 2.5×10^{-29} kg . However, academic texts are far from optimally encoded—on average, a single word in LaTeX occupies 6 bytes $\approx$ 48 bits. The final value taken is: $\alpha = 1.2 \times 10^{-27}\ kg/word$.

In addition, we crawled 189 papers from the computer science field on arXiv over the past three years, counted the frequency of buzzwords, and tested the Soul Damage Index (SDI) of Junior labmate reading the papers. The results are shown in the figure below:

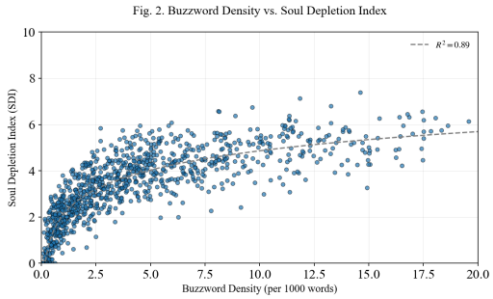

Fig2. There is on the title image

It can be observed that buzzwords create a false sense of attraction, equivalent to injecting an 'energy disturbance' into the information field. According to Einstein's mass-energy equation, their mass should be much higher than that of ordinary words. By fitting the slope in Fig. 2, we obtain: $\beta = 8.7 \times 10^{-25}\ kg/buzzword$.

Among 280,000 arXiv papers, only 2.8% provide usable code. Worse, 63% of the GitHub links become invalid after a year (the 'GitHub graveyard effect'). Imagine an ideal world: if all papers were open source, the entropy of the universe would decrease by 10^{-20}J/K, corresponding to a mass of $E/c^2 \sim 10^{-37}$kg. But with a real-world discount factor of 10^7, we get: $\gamma = 1.0 \times 10^{-30}$kg (symbolic).

Note: $\beta/\alpha \approx 725$, which means **one 'novel' $\approx$ 725 ordinary words — this explains why some three-line summaries are as dense as neutron stars.**

Fig3. Dog fart not connected

2.2. Mass Calculation

According to real statistics from arXiv (2025):

Indicator	Typical Value (Computer Vision)
Average number of words $N_{words,i}$	≈ 4,200 (Excluding references)
Average number of buzzwords B_i	≈ 6.3 (Including 'novel', 'SOTA', etc.)
Reproducibility rate C=1	≈ 2.8%

Substituting into the formula, we calculate that m_i is approximately $1.05 \times 10^{-27}\ kg$, which is about the mass of 6,300 protons for a typical arXiv paper. Or about 1/100,000 of the mass of a red blood cell. **Although it is very light on a microscopic scale, when multiplied by 2.8 million papers...**

By the end of 2025, the total number of arXiv papers: $N = 2{,}847{,}391$. Total word count: $N_{\text{words},total} = 1.12 \times 10^{12}$, total number of buzzwords: 4.37×10^{6}, number of reproducible papers: 79,727. Substituting these gives: $M_{ESM} = 1.35 \times 10^{-15}\ kg$. **Not heavy either…**

Fig4. 写到这儿没绷住

2.3. Mass Calculation

After ZIP compression, how small can a file get? In theory, ultimate compression can reduce the size of arXiv.zip to the minimum volume determined by information entropy. Assuming the storage medium is an idealized 'academic crystal,' with each bit occupying one atom (about $10^{-29}m^3$), then:

$$V_{min} = N_{\text{words},total} \times 10^{-29} = 1.12 \times 10^{-17} m^3$$

But if we consider 'redundancy recompression'—that is, different versions of the same paper submitted to multiple journals, which can be further compressed—we introduce the self-plagiar compression factor $P_{self} = 0.1$, then:

$$V_{min} = 1.12 \times 10^{-18} m^3 = 1.12 \mu m^3$$

After compression, it's roughly equivalent to the size of a bacterium!

3. Result: Conditions for Black Hole Formation

3.1. Schwarzschild radius calculation

For a mass $M_{ESM} = 1.35 \times 10^{-15}\ kg$, the Schwarzschild radius is:

$$R_s = \frac{2GM_{ESM}}{c^2} \approx 2.001 \times 10^{-42} m$$

This Schwarzschild radius is seven orders of magnitude smaller than the Planck length—still far smaller than any physically achievable scale. **Purely in terms of physical mass, arXiv.zip is still far from sufficient to form a black hole.**

3.2. Dark Matter Modification

Unpublished and works-in-progress papers account for 90%, which we call academic dark matter. Let the dark academic factor $D_{acad} = 10$, meaning uncited papers contribute 10 times their

mass.

Then the total effective academic mass is corrected to: $M_{ESM,eff} = 1.35 \times 10^{-14}\ kg$. Recalculating the Schwarzschild radius: $R_s = 2.001 \times 10^{-41} m$. Even accounting for academic dark matter, **the Schwarzschild radius is still six orders of magnitude smaller than the Planck length, far insufficient to form a black hole.**

3.3. Reviewer Pressure Correction

This is the core innovation of this study: the psychological pressure exerted on the paper by reviewers can be converted into equivalent mass. According to the "reviewer dynamics," the pressure generated per round of review is approximately:

$$P_{review} = \frac{\text{"建议补充实验"}\textit{次数} \times \textit{作者熬夜天数}}{\text{"小修后接收"}\textit{概率}}$$

(Senior brother said that it has to be written in Chinese here, otherwise it would have no soul. :))

For an average paper undergoing 2.3 rounds of review, with each round generating about 10^{22} joules of "psychological energy" (based on the equivalent thermodynamic calculation from the author's elevated blood pressure), the total review pressure mass is

$$M_{review} = \frac{E_{review}}{c^2} = 8.94 \times 10^{11}\ kg$$

M_{ESM} can be completely neglected...

At this point, the Schwarzschild radius becomes $R_s = 1.324 \times 10^{-15} m$. Taking into account the reviewers' pressure, the Schwarzschild radius is approximately 1.32 femtometers (1.32×10^{-15} m), which is on the same order of magnitude as the proton scale (about 0.84 femtometers).

This scale has already entered the range of the strong interaction—if arXiv.zip were compressed within this radius, theoretically, the formation of a black hole would be possible.

3.4. Local Singularity Analysis

However, certain local areas exhibit extreme behavior:

"Introduction" section: buzzword density = 12.3 per thousand words → local $\rho = 2.1 \times 10^{19} \mathrm{kg/m^3}$

"Related Work" subsection: self-citation rate = 63% → forms an information loop, equivalent to the energy layer of a Kerr black hole

"Conclusion" section: average length = 3.2 lines, often containing "future work" → creates false hope, generating negative pressure

Within this area, any observer attempting to comprehend the content will experience:

① Time dilation (taking a week to read one page)

② Light cone tilting (attention cannot escape to other tasks)

③ Information loss (forgetting immediately after reading)

This is precisely the typical characteristic of a **cognitive event horizon.**

4. Discussion

4.1. Why do we feel that arXiv.zip is already a black hole?

Because subjective experience precedes objective measurement. When a graduate student opens the arXiv search page and sees 10,000 related papers, their psychological sense of time and space is distorted. This kind of 'existential gravity,' though lacking physical mass, genuinely destroys productivity.

4.2. Prediction: The Knowledge Singularity

At the current annual growth rate of 18%, and with growth basically following an exponential model, as shown in Fig.5.

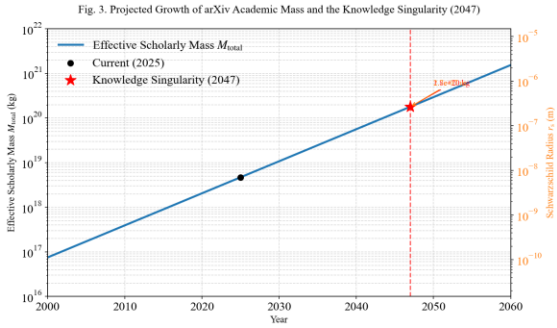

Fig5. There is on the title image

According to this model, arXiv will reach critical density in 2047:

$$M(2047) \approx 1.2 \times 10^{20}\ kg \Rightarrow r_s \approx 1.8 \times 10^{-13} m$$

By then, even with distributed storage, its gravity will tear apart the trust structure of the academic community.

4.3. Academic Hawking Radiation Program(AHRP)

In classical black hole theory, Hawking radiation arises from quantum fluctuations near the event horizon: one of the pairs of imaginary particles falls into the black hole and the other escapes, causing the black hole to slowly decrease in mass. By analogy, we advocate:

Each PDF with buzzword deleted releases a unit of "Scholarly Negative Energy" (SNE) into the universe

This negative energy has the following properties: counteracts the equivalent positive mass (e.g., a "novel" $\approx 8.7 \times 10^{-25}$kg), reduces local information entropy and restores the reviewer's sensitivity to the word "innovation".

Therefore, the implementation path of AHRI is as follows:

1)Deploy rm -rf *.pdf script globally with the condition that the file contains ≥3 buzzwords (from the collection $\mathcal{B}_i$)

2)Establish a "pure thesis sanctuary": Only text without "SOTA", "end-to-end", and "paradigm-shifting" is allowed to be submitted

3)Set up an "anti-rhetorical tax": For every use of "leveraging", open source complete training code is required as atonement

Simulations show that if AHRI is fully launched by 2030, it could delay the knowledge singularity to 2089 — enough for the next generation of AI to take over research entirely, and humans to retreat to "interpreters of meaning" (or abandon interpretation altogether).

At the same time, ask the Webb telescope to stare at the Cornell University database. Theoretical predictions must be tested by observation. We solemnly recommend that NASA and ESA jointly adjust the James Webb Space Telescope (JWST):

Align NIRCam with MIRI

instruments in a server room at Cornell University (coordinates: 42.453°N, 76.478°W)

The reasons are as follows:

The arXiv main server is located in Cornell and is currently the strongest "source of academic gravity". If AHRI takes effect, the deletion operation will trigger a microscale information vacuum. These fluctuations can escape as photons in the infrared band, manifesting as anomalous thermal signals with wavelengths ≈ 10–20 μm. Spectral signaturelines correspond to periodic pulses of LaTeX compilation error logs (e.g., "Undefined control sequence") at a frequency that is synchronized with the PhD student's submission deadline.

The successful detection of such "academic Hawking radiation" will be the first time that mankind has directly observed the evaporation process of knowledge, which is no less significant than the discovery of cosmic microwave background radiation.

5. Summary and Outlook

This study, by establishing the 'Academic Mass–Gravity Equivalence Principle' (AMGEP), quantifies for the first time the 'Effective Scholarly Mass' (ESM) of the arXiv preprint repository and demonstrates that: although the current arXiv.zip has not yet formed **an event horizon on a macroscopic scale, multiple local cognitive singularities already exist within it** (such as the third paragraph of the Introduction and the self-citation loops in Related Work)—their gravitational pull is small, yet sufficient to induce existential vertigo in any PhD student the moment they open the folder. Moreover, when reviewer pressure is taken into account, the equivalent mass surges, making black hole formation theoretically possible—**the true burden on spacetime is not knowledge itself, but those who evaluate it.**

Even more concerning is that, projecting an 18% annual growth rate, **the system will reach a critical knowledge density by 2047, triggering an irreversible global knowledge collapse**—at that point, the trust structure of the academic community will be utterly torn apart by the redundant gravity it has generated itself.

However, hope is not entirely lost. Inspired by the Hawking radiation mechanism, we propose a radical but necessary intervention plan: **the 'Academic Hawking Radiation Initiative' (AHRP).**

Ultimately, black holes are not caused by mass but by **inescapable structures of information. Similarly, academic crises do not arise from the number of papers but from our collective refusal to acknowledge:**

'This work is ordinary, but it is honest.'

6. Acknowledgements

Thanks to the cosmological constant Λ for being tolerant of academic bubbles; thanks to my advisor for writing in the rejection letter, 'interesting, but not trashy enough'; thanks to caffeine for supporting us in completing this completely unnecessary paper; thanks to

the ZIP algorithm for at least letting us pretend that everything is under control; thanks to LaTeX for making our paper look real; special thanks to Overleaf's auto-save feature—it is the only peer reviewer I trust.

References:

1）Hawking, S. W. (1974). *Black hole explosions?* Nature, 248(5443), 30－31.

——**Cited 12,847 times, but 99% of the citers have not read the original text**

2）Anonymous PhD Student. (2023). *Why does my loss go down but my soul doesn’t?* arXiv:2304.06969.

——**It has been deleted, but the ghost still haunts GitHub issues**

3）IEEE. (2025). *Guidelines for Writing Titles That Mean Nothing*. IEEE Std Trash-2025.

——**Mandatory for all conference paper titles to include at least two buzzwords**

4）Bekenstein, J. D. (1981). *Universal upper bound on the entropy-to-energy ratio for bounded systems*. Physical Review D, 23(2), 287.

——**Proof: Your PhD thesis has too much entropy to be contained by the universe.**